



CSCI 3052U: Machine Learning 1

Milestone 1: Proposal And Team Contract

Github Link:

<https://github.com/kandoimansi/Machine-Learning-Project-Group-18.git>

GROUP 18

Team Members

Aryan Patel: 100917325

Mansi Kandoi: 100821953

Sadhna Drall: 100885616

1. Title

Credit Card Fraud Detection Using Machine Learning Techniques

2. Abstract

Credit card fraud is a major challenge for consumers and financial institutions, particularly with the growth of online transactions. Traditional rule-based systems often struggle to detect evolving fraud patterns effectively. This project aims to develop a machine learning-based approach for identifying fraudulent credit card transactions using a publicly available dataset. A majority-class baseline will be compared with Logistic Regression, Decision Tree, and Random Forest models. Data preprocessing techniques such as normalization and class imbalance handling will be applied to improve performance. The models will be evaluated using Precision, Recall, F1-Score, and ROC-AUC, with emphasis on Recall due to the high cost of missed fraud cases. The project seeks to identify an effective fraud detection model and demonstrate the value of machine learning in improving financial security.

3. Research Background

Problem Being Investigated - Credit card fraud occurs when unauthorized individuals use a person's credit card information to perform transactions without permission. As digital banking and online shopping continue to grow, financial institutions face increasing challenges in identifying fraudulent transactions quickly and accurately.

Current Understanding - Traditional fraud detection relies on rule-based systems that flag suspicious transactions using predefined conditions. However, these systems require regular updates and often fail to adapt to new fraud patterns. Machine learning provides a data-driven alternative that can automatically learn complex patterns from historical transaction data.

Research Gap - Despite improvements in fraud detection, challenges remain due to highly imbalanced datasets and the difficulty of detecting fraud while minimizing false positives. This project aims to compare multiple machine learning models and identify the most effective approach for fraud detection.

Importance of the Research - Improving fraud detection can reduce financial losses, improve customer trust, and enhance the security of online transactions. This project also demonstrates the application of machine learning techniques to a real-world problem.

4. Methodology

Data - The project will use a publicly available credit card fraud detection dataset obtained from Kaggle. The dataset contains transaction records labeled as either legitimate or fraudulent.

The dataset will be analyzed to understand: Number of transactions, Class distribution, Transaction features, Missing values and data quality issues.

Methods

The following machine learning approaches will be implemented and compared:

Baseline Model - Majority-class prediction (classifying all transactions as legitimate)

Machine Learning Models - Logistic Regression, Decision Tree, Random Forest

Data preprocessing may include: Data normalization, Feature selection, Handling class imbalance, Train-validation-test splitting

Experiments and Evaluation - All models will be trained and tested using the same dataset split to ensure a fair comparison. Performance, fraud detection capability, and computational efficiency will be evaluated.

Evaluation Metrics - Performance will be measured using: Precision, Recall, F1-Score and ROC-AUC. Recall will be considered particularly important because failing to identify fraudulent transactions can be more costly than incorrectly flagging legitimate ones.

Comparison - The performance of Logistic Regression, Decision Tree, and Random Forest models will be compared against: the majority-class baseline and each other under identical conditions. The best-performing model will be identified based on overall effectiveness and reliability.

5. Significance

This project contributes to the use of machine learning in fraud prevention by comparing multiple classification techniques on an imbalanced fraud dataset. The findings may benefit financial institutions, credit card companies, and consumers by improving fraud detection systems. The results may also support future research in advanced fraud detection and financial risk analysis.

6. SMART Timeline

Time Period	Milestone	Tasks
Sept 19 - Oct 2, 2026	Milestone 2: Dataset and Problem Approval	Obtain credit card fraud dataset, perform exploratory data analysis (EDA), document data source and license, define evaluation metrics, establish train/validation/test split strategy, and assess feasibility.
Oct 3 - Oct 19, 2026	Milestone 3: Baseline Reproduction	Implement data preprocessing pipeline, address class imbalance, create a naive baseline model, implement Logistic Regression baseline, document reproducibility procedures, and evaluate baseline performance.
Oct 20 - Oct 30, 2026	Milestone 4: Progress Presentation	Train Decision Tree and Random Forest models, generate preliminary results, compare model performance, prepare progress presentation, and revise project plan based on feedback.
Oct 31 - Nov 20, 2026	Milestone 5: Final Experiments	Conduct hyperparameter tuning,

	and Analysis	perform final model comparisons, evaluate Precision, Recall, F1-Score, and ROC-AUC, conduct error analysis, and document findings.
Nov 21 - Dec 4, 2026	Milestone 6: Final Report, Repository, and Demonstration	Finalize analysis and conclusions, complete repository documentation, prepare presentation slides, write final report, ensure reproducibility, and submit final project deliverables.

7. References (IEEE Style)

- [1] A. Dal Pozzolo, O. Caelen, Y. A. Le Borgne, S. Waterschoot, and G. Bontempi, "Calibrating Probability with Undersampling for Unbalanced Classification," in IEEE Symposium Series on Computational Intelligence (SSCI), Cape Town, South Africa, 2015, pp. 159-166.
- [2] MLG-ULB, "Credit Card Fraud Detection Dataset," Kaggle. [Online]. Available: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud> [Accessed: Sept. 20, 2026].
- [3] S. Bhattacharyya, S. Jha, K. Tharakunnel, and J. C. Westland, "Data Mining for Credit Card Fraud: A Comparative Study," Decision Support Systems, vol. 50, no. 3, pp. 602-613, Feb. 2011.